Better Health = Stronger Communities

HealthBridge

Community Well-Being & Lifestyle Tracker

Small steps make a big impact

Project Documentation

Track Your Health

Build Better Habits

See Your Progress

Stronger Communities



Aligned with UN SDGs

3 GOOD HEALTH AND WELL-BEING

10 REDUCED INEQUALITIES

Health for all, no one left behind.

Healthier People
Happier Communities
A Brighter Future ♡

Inside This Report

- Objectives & Overview
- System Design & Architecture
- OOP Concepts
- Algorithms & Logic
- SDG Mapping
- Input/Output Flow
- Testing & Limitations
- Future Scope

And much more...!

Code
Ideas
Impact



Documentation
Swarit & Yunay

</> Coding & Demonstration
Jivraj & Vaibhav

Presentation
Venisha & Pradishan

Team ☺

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HEALTHBRIDGE

PROJECT FIELD GUIDE / Everyday well-being, made easier to understand

01 The idea and the problem

A quick note before you begin

This guide follows HealthBridge from its interface to the logic behind it. It is designed to help readers understand the app, explain its features in a presentation, and recognise its current boundaries.

WHAT IS HEALTHBRIDGE?

Health Bridge is a simple, zero-cost web dashboard designed to help people manage their daily health habits. Instead of waiting until you get sick to take action, it acts as a preventative tool to help you stay healthy every day. The app brings basic daily habits—like sleep, steps, water intake, mood, and stress—into one clean interface. It uses a mathematical engine to turn those daily numbers into a single, easy-to-read Composite Wellness Score out of 100. Unlike mainstream fitness apps that push expensive gym memberships, organic meal plans, or high-end smartwatches, Health Bridge is built for everyone. It focuses on free, barrier-free habits, such as drinking tap water or taking a short walk after dinner. This aligns the project with SDG 3 and SDG 10.

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The problem in one sentence

Daily entries can be difficult to interpret on their own. HealthBridge brings recent habits together in a compact, understandable view so users can reflect on patterns and decide what they may want to change.

02 Project objectives and boundaries

- **Centralise health tracking:** Bring sleep, activity, hydration, mood and stress into one dashboard.
- **Support prevention:** Use scores and trends to highlight changes in everyday habits.
- **Reduce cost barriers:** Focus on free activities and simple lifestyle metrics that do not require expensive equipment.
- **Make results actionable:** Pair a score out of 100 with clear, non-medical suggestions.
- **Protect data quality:** Apply server-side validation and organised OOP design.
- **Enable portability:** Generate reports in CSV and JSON formats.

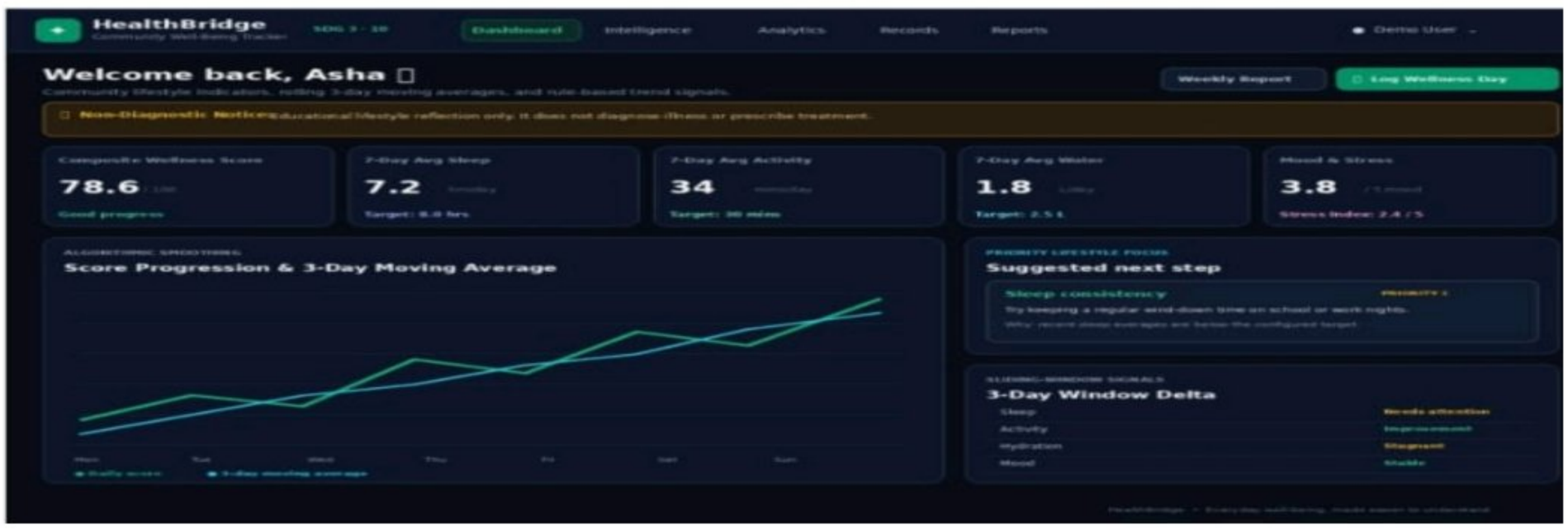
What success looks like

For a classroom demonstration, success means a person can add a valid record, see it reflected in the dashboard, understand why a score or suggestion appeared, and explain the system's boundaries honestly.

03 Features and screen-by-screen tour

Screen / feature	What it does
Interface	A clean, readable layout with explanatory text to help users understand each feature.
Chatbot / Community Coach	A basic assistant answers supported questions about trends, recommendations and scores. The community concept focuses on free lifestyle tips.
Dashboard	Shows the wellness score, seven-day metric averages and a priority lifestyle focus for the day.
Intelligence Lab	Explains score weights, bonuses and penalties; a What-If predictor lets users test habit changes.
Analytics Charts	Shows multi-indicator trends, a five-pillar balance view and percentage fulfilment bars.
Records and Daily Log	Displays historical entries and lets users enter daily sleep, activity, water, mood and stress values.
Validation Lab	Demonstrates how invalid or corrupted entries are blocked by server-side validation.

The main dashboard



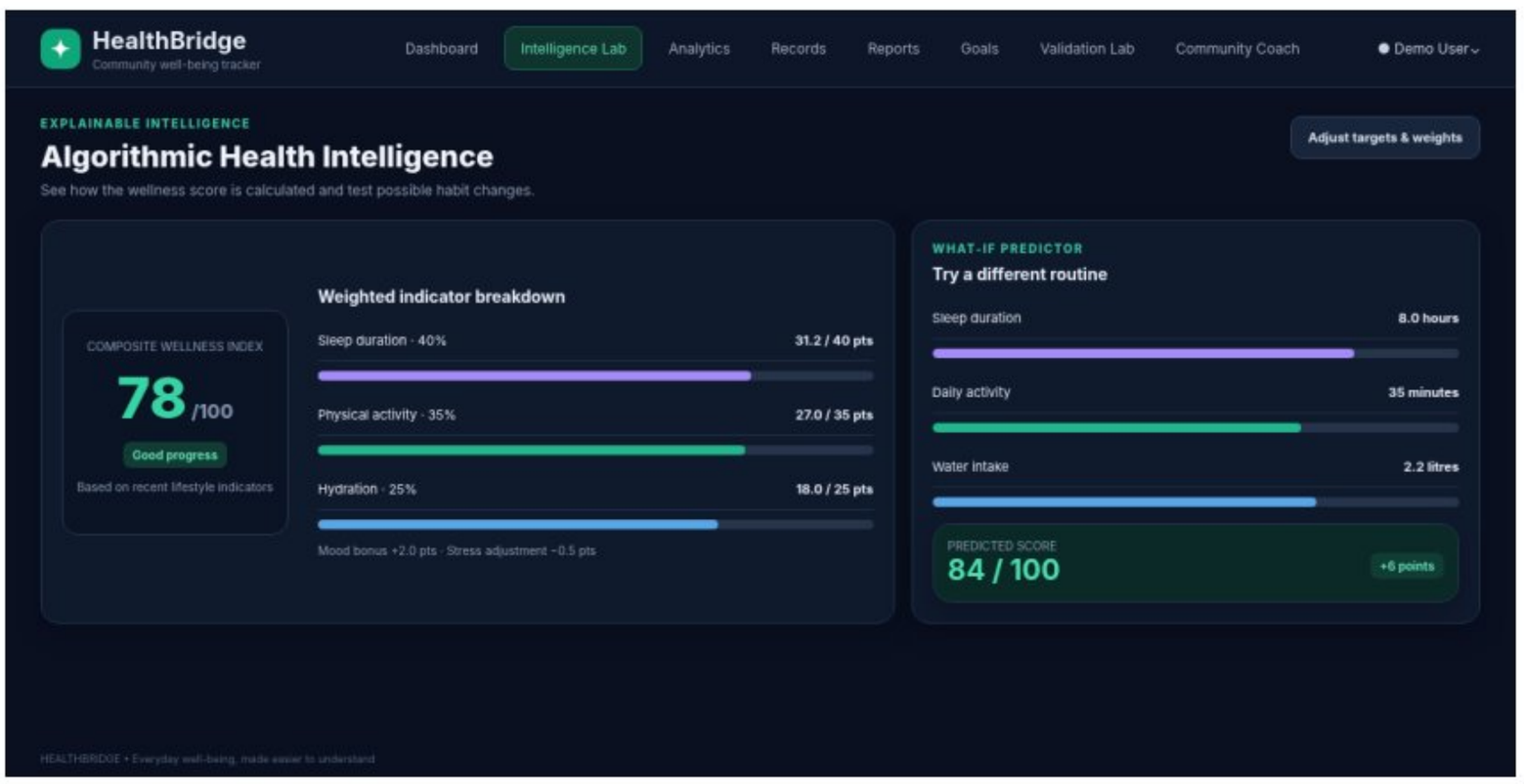
The dashboard puts the main indicators first. Snapshot cards make averages easy to scan, trend charts show movement across days, and the priority card suggests a practical next step. Values shown in this recreated dashboard are illustrative placeholders.

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03 Features and screen-by-screen tour

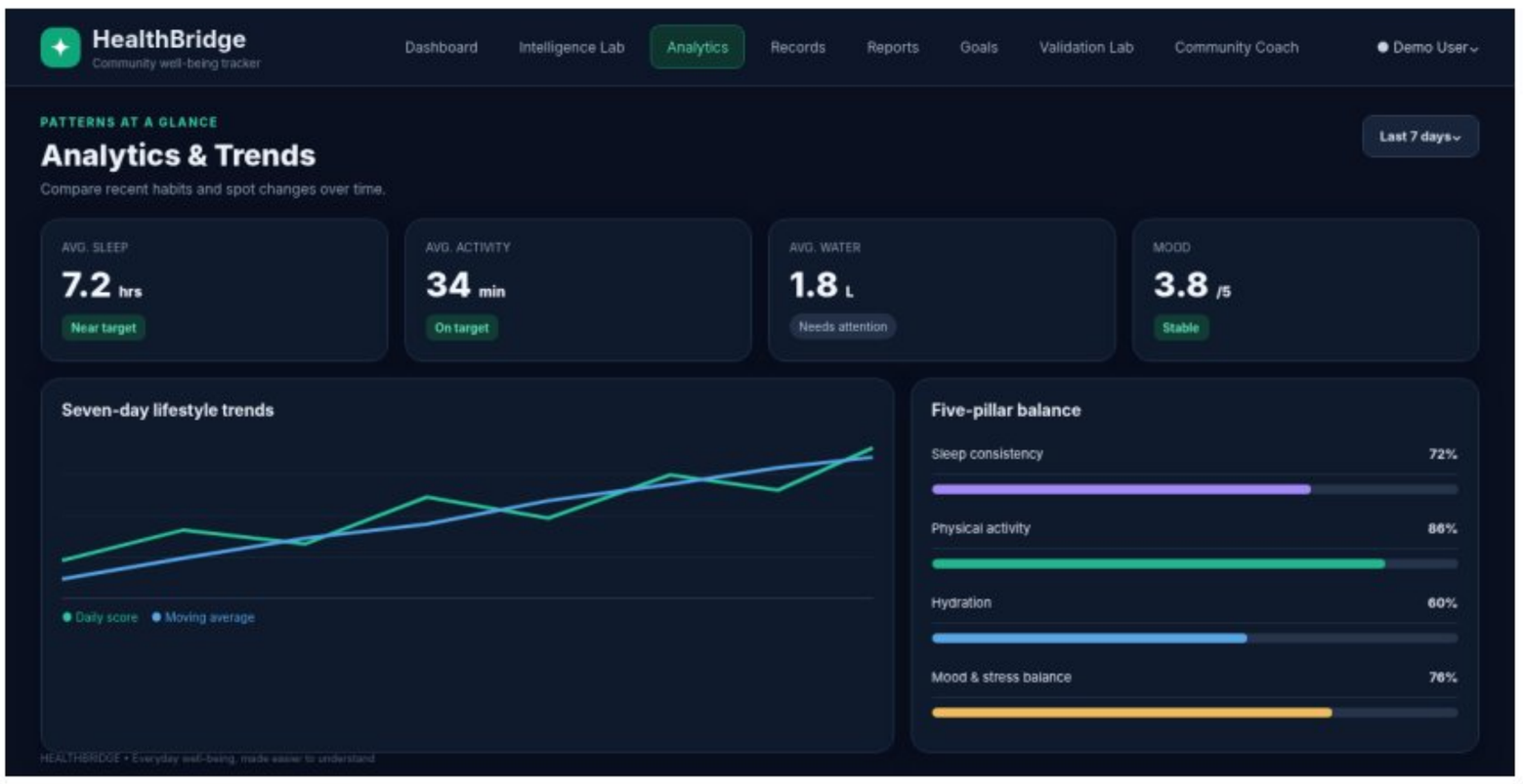
Application interface screenshots · illustrative demo data

Intelligence Lab



The Intelligence Lab explains how the composite wellness score is formed from recent lifestyle indicators. Its weighted breakdown makes the contribution of sleep, activity and hydration easier to understand. The What-If panel illustrates how changing example habits can affect a projected score; these values are demonstrations, not medical predictions.

Analytics Charts



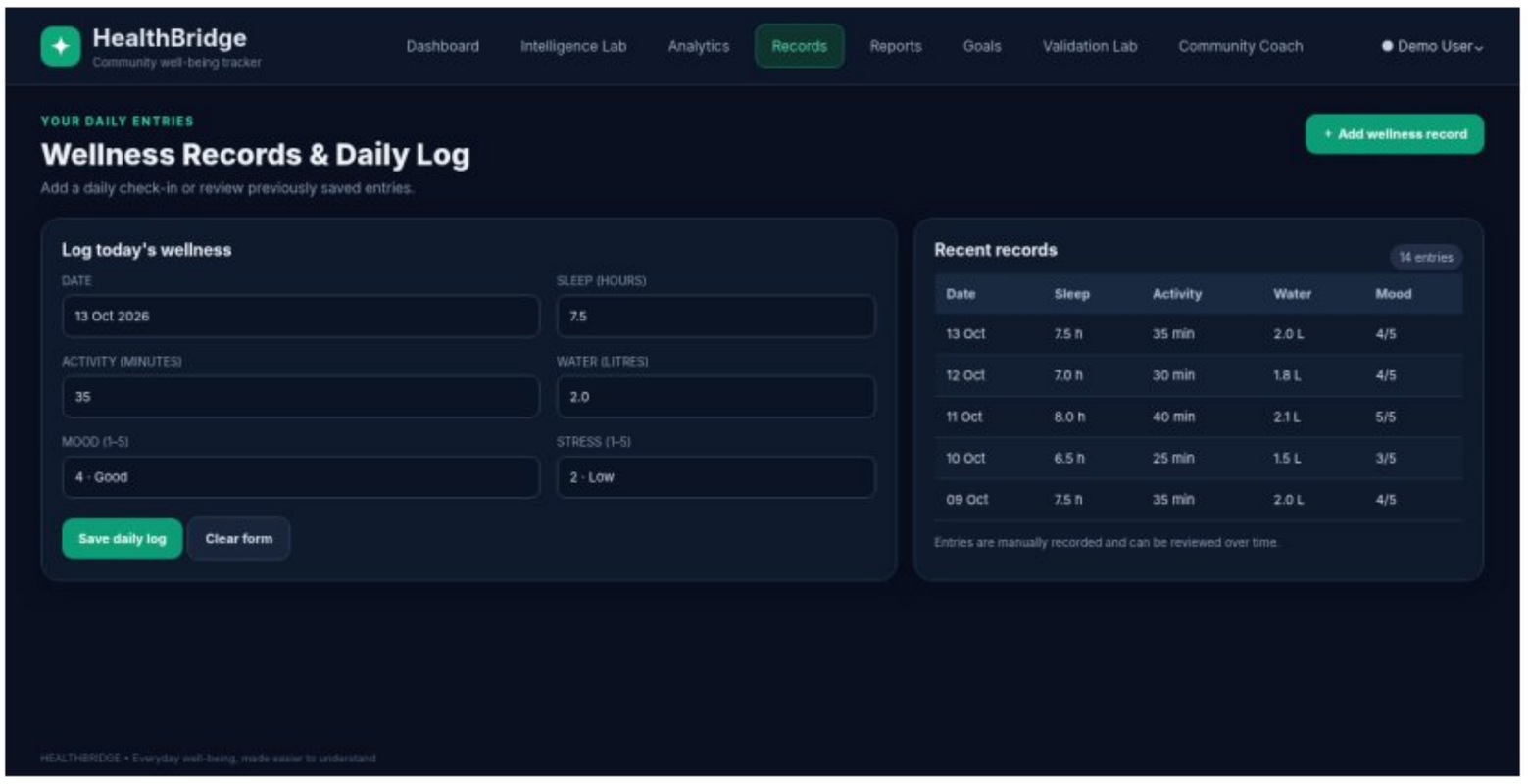
The Analytics screen brings recent indicators together so users can compare patterns over a selected period. Summary cards show average sleep, activity, water intake and mood, while the charts help reveal changes over time. The displayed figures are sample values for explaining the interface.

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Feature screens

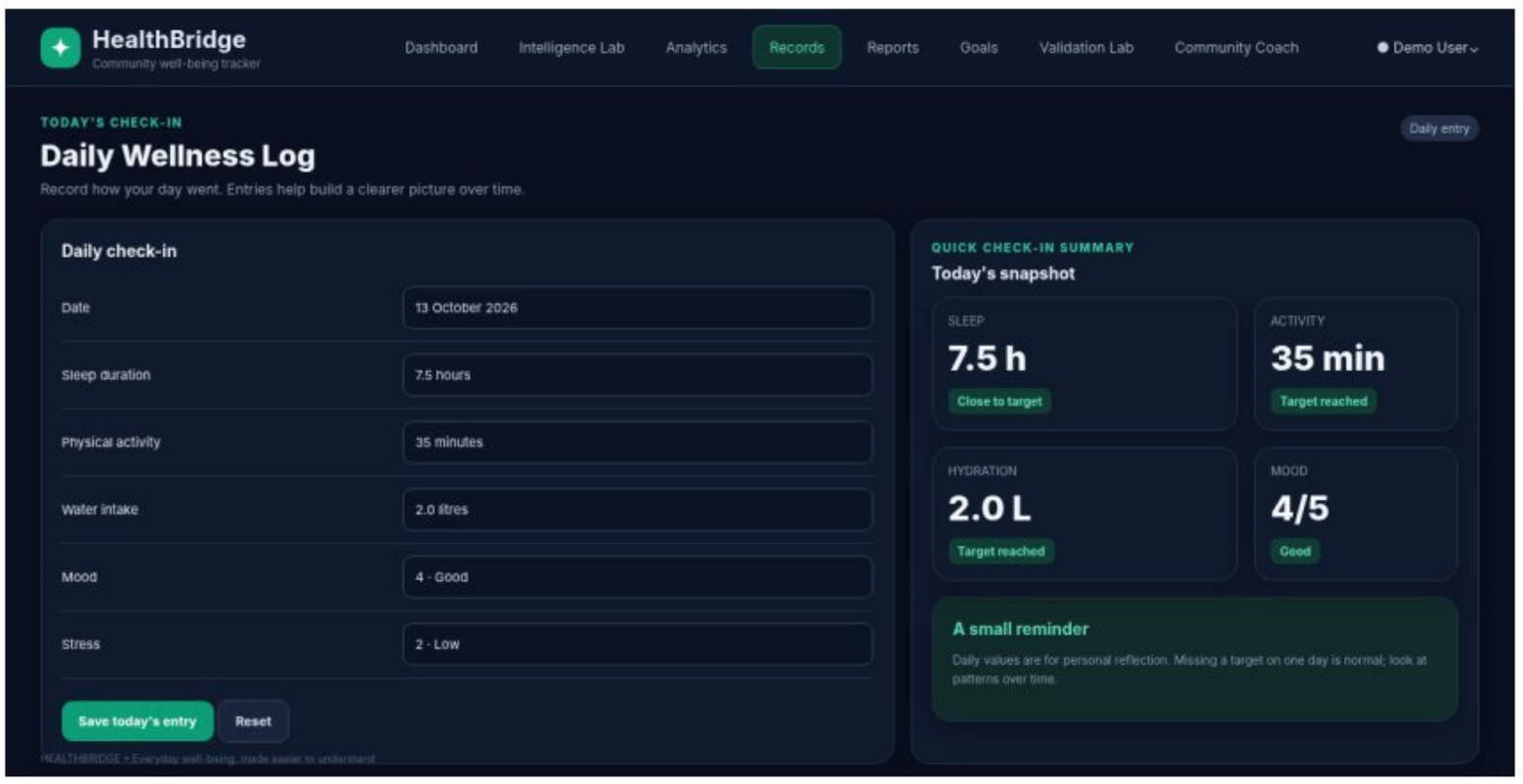
Screen descriptions and interface walkthrough

Records



Records is the history view for previously saved wellness entries. The table lets users review dates and recorded indicators together, making it easier to compare one day with another and check what has already been entered. The entries shown here are illustrative demo data.

Daily Logs



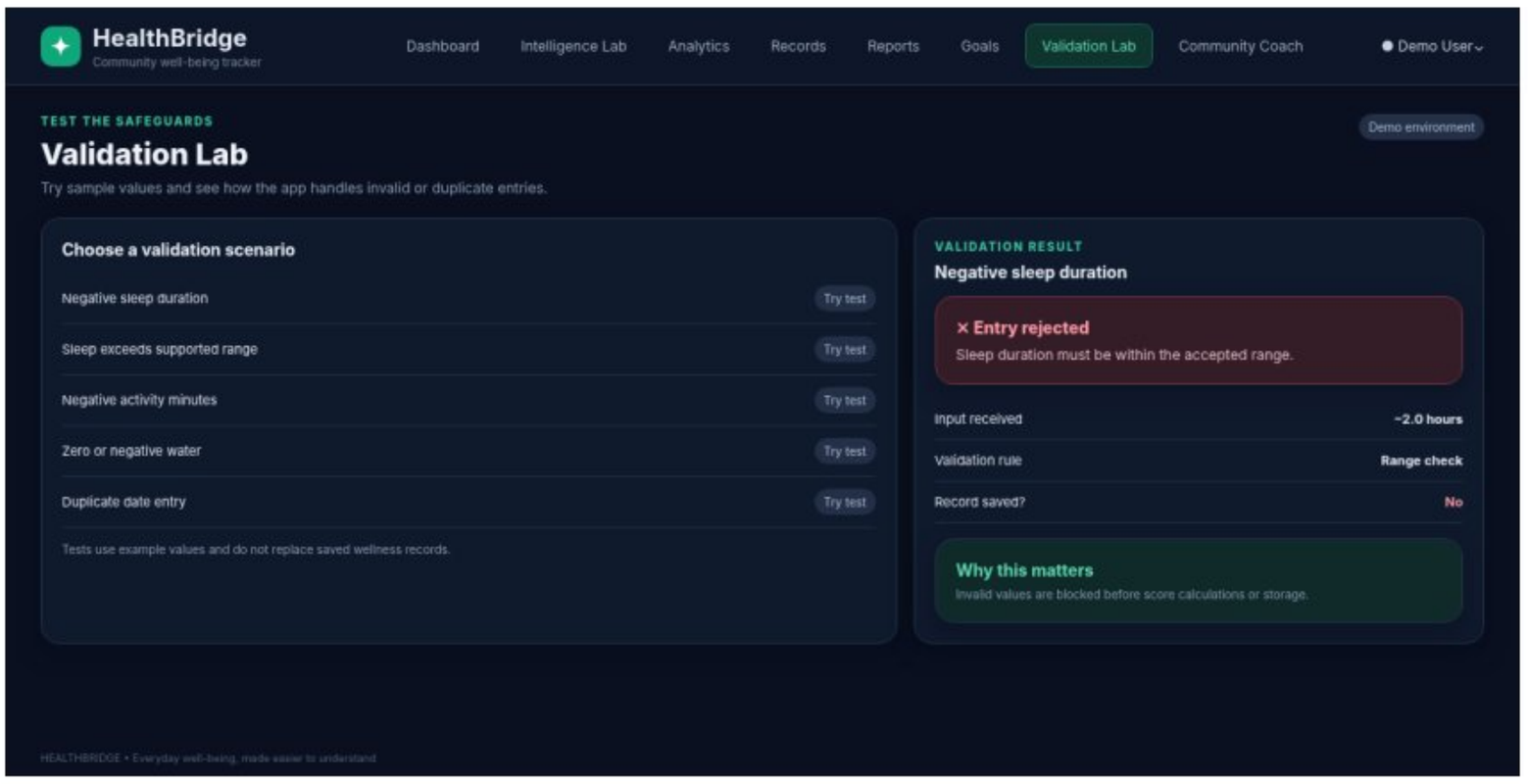
Daily Logs is the check-in screen for entering a day's wellness information, such as sleep duration, physical activity, water intake, mood and stress. The summary panel provides a quick view of the entered values before or after saving. The example screen demonstrates the form and does not represent a real person's health data.

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Feature screens

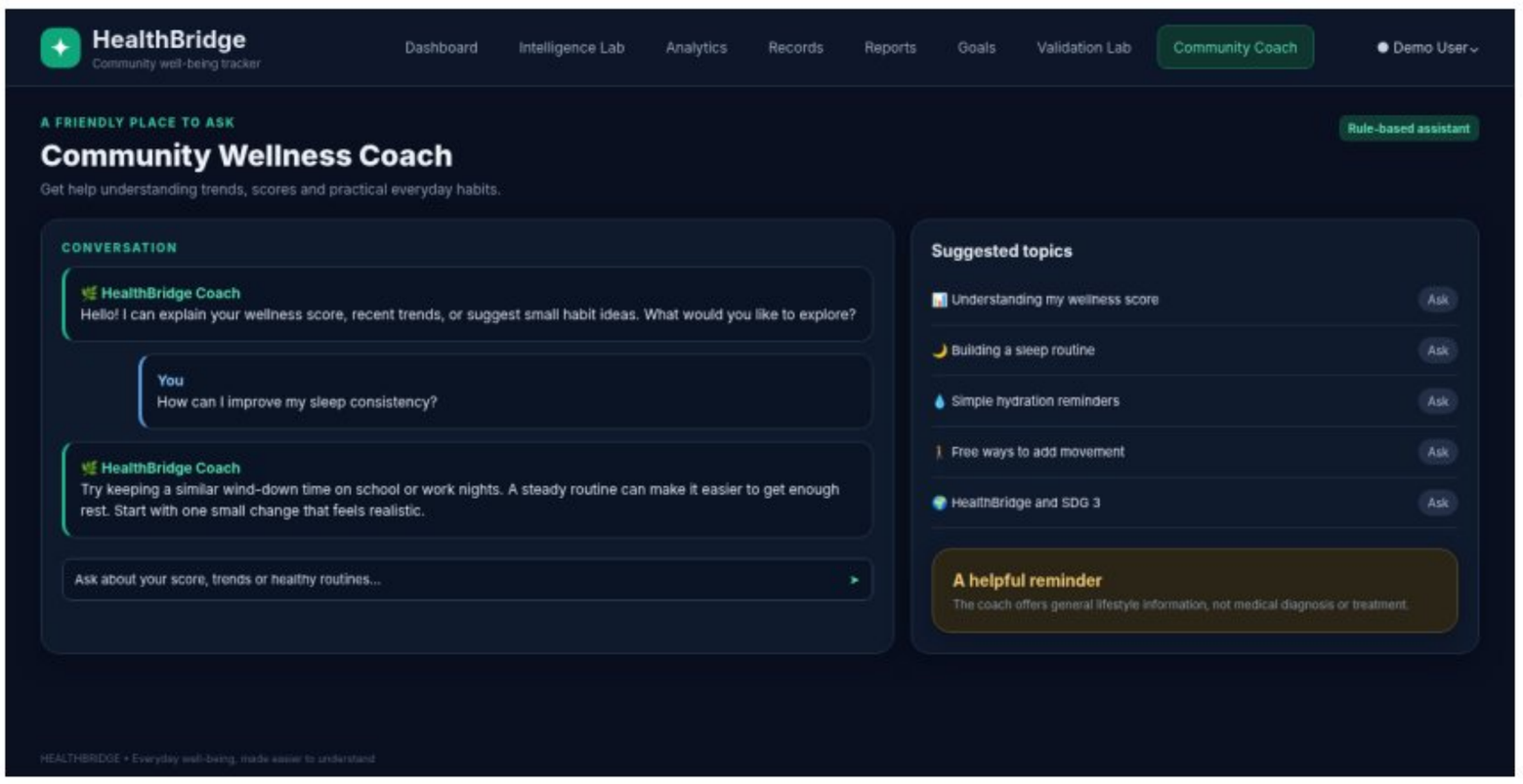
Screen descriptions and interface walkthrough

Validation Lab



The Validation Lab demonstrates safeguards that check input before it is used. Example scenarios include negative or out-of-range values and duplicate dates. The result panel explains why an entry may be rejected, helping show how invalid data is kept from affecting records and calculations.

Community Coach



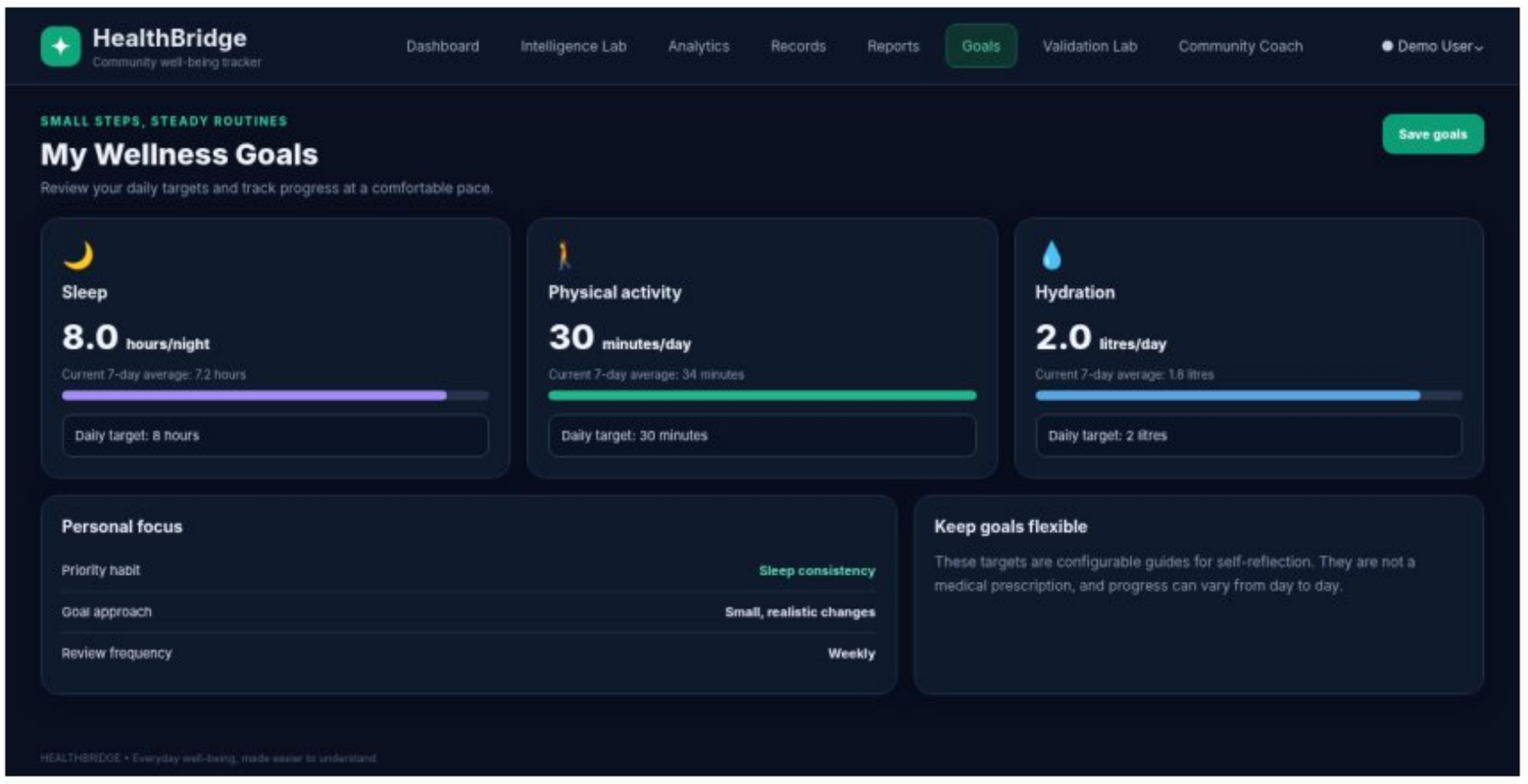
Community Coach provides a conversational way to explore general wellness topics, understand indicators and find small habit ideas. Suggested prompts help users get started. It is intended for general lifestyle information and reflection, not diagnosis, treatment or emergency advice.

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Feature screens

Screen descriptions and interface walkthrough

Goals

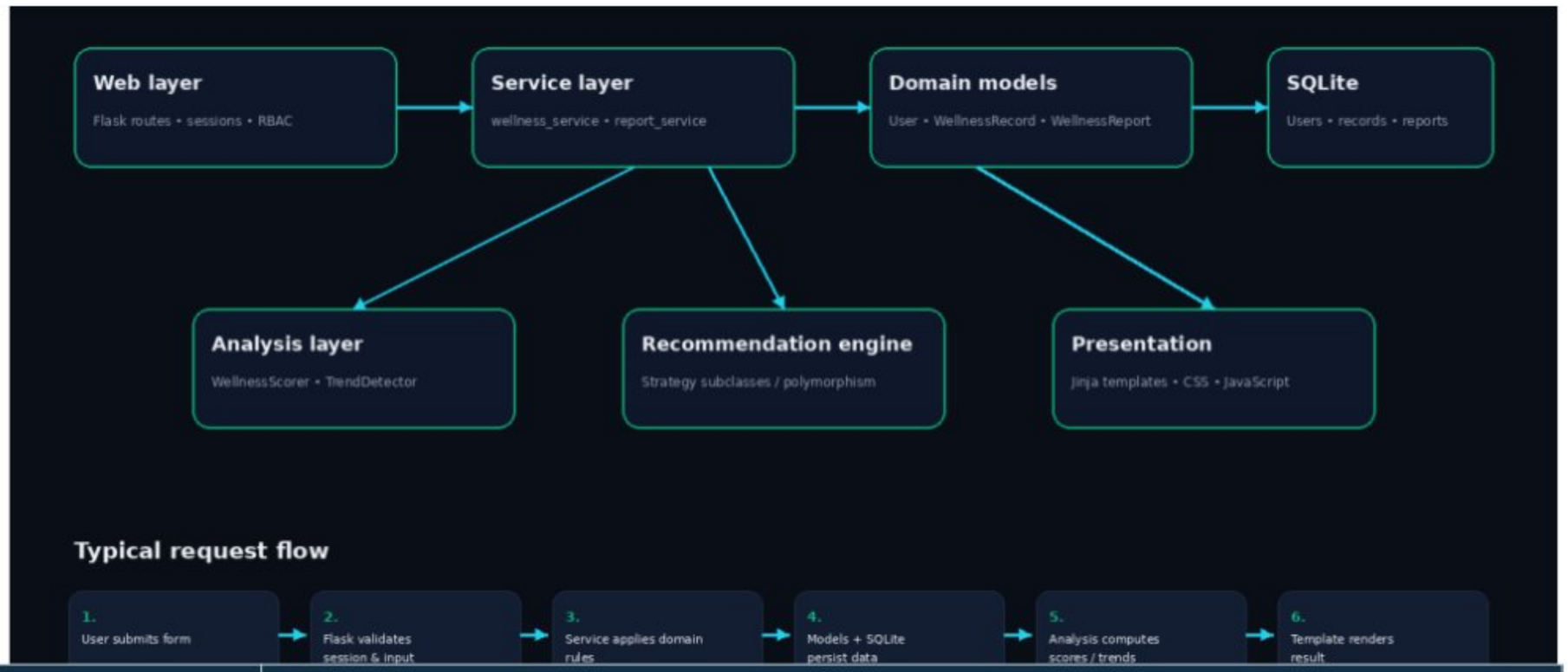


The Goals screen displays configurable targets for sleep, physical activity and hydration alongside recent averages and progress bars. This helps users compare their current routine with the targets they have chosen and identify a manageable focus. Targets are guides for self-reflection, not medical prescriptions.

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04 What happens behind the screen

The application is organised in layers so that browser routes, data rules, analysis and storage do not all have to do the same job. This separation makes the project easier to explain, test and maintain.



Layer	Responsibility
Presentation	Jinja templates and static CSS/JavaScript render the interface.
Web / route layer	Flask handles requests, sessions and route-level access checks.
Service layer	Coordinates record workflows and report generation.
Domain models	Represent users, daily records and reports, with relevant rules.
Analysis layer	Calculates scores, compares trends and selects suggestions.
Persistence	SQLite stores application data; exports provide CSV or JSON output.

Typical request flow

- A user opens a page or submits a form.
- Flask checks the session and permissions.
- The service and model layers validate the entry; accepted data is stored.
- Analysis calculates scores, trends or suggestions when needed.
- Flask returns a rendered page or export.

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05 OOP: the classes and why they matter

A typical request, step by step

1. A user opens a page or submits a form.
2. Flask receives the request and checks the session and required permissions.
3. The service layer validates and coordinates the operation.
4. The model rules accept or reject the record; accepted data is stored.
5. Analysis code calculates averages, a score, trends or suggestions when needed.
6. Flask returns a rendered page or an export response.

Object-Oriented Design in HealthBridge

Encapsulation (data hiding)

WellnessRecord wraps health data and validation rules so that records remain consistent.

Abstraction (standard templates)

An abstract base class provides a shared contract so related components follow a consistent structure.

Single responsibility

Each class has a focused role: UserModel manages user profiles, WellnessRecord stores health values, and engine modules calculate scores.

Why this design helps

Clear responsibilities make the code easier to locate, test and extend without changing unrelated parts.

06 Algorithms & Logic Breakdown

Validation before calculation

The app checks dates and supported numeric ranges before accepting an entry. It also prevents duplicate records for the same date.

Composite wellness score

The core score uses configured targets of 8 hours of sleep, 30 minutes of activity and 2 litres of water. Sleep contributes 40%, activity 35% and hydration 25%; each core indicator is capped at 100 before weighting. Mood may add a bonus and stress may apply a penalty. The final score is a configurable summary, not a clinical measurement.

Trend detection and suggestions

TrendDetector compares a recent three-day window with an earlier period. Larger changes may be labelled improvement or decline, while small changes can be treated as stable. Recommendation rules compare recent averages with targets and trends to suggest practical, non-medical next steps.

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Operation	Approximate cost	Reason
Validate one record	O(1)	A fixed set of fields is checked.
Find duplicate date	O(n)	The records may need to be scanned.
Sort records	O(n log n)	Sorting grows with list size.
Calculate window average	O(n)	Each record in the window is examined.

07 Data journey: input to output

Stage	Example	What comes next
Input	Date, sleep, activity, water, mood/stress	The route receives form values.
Validation	Format, ranges and duplicate-date check	Invalid input receives feedback.
Save	Accepted record stored through the model/database path	It becomes available for later views.
Analyse	Recent records feed averages, score and trends	Structured indicators are produced.
Recommend	Rules compare indicators with targets and trends	Relevant suggestions are selected.
Output	Dashboard, report or CSV/JSON export	The user can review the summary.

08 SDG mapping: connecting the project to a bigger picture

HealthBridge connects most directly to SDG 3, Good Health and Well-Being, and also to SDG 10, Reduced Inequalities. The project can support awareness and make information easier to access, but it should not claim measured public-health impact without evidence.

UN Sustainable Development Goal	Connection to the project	Responsible claim
SDG 3 · Good Health and Well-Being	Makes everyday patterns around sleep, movement, hydration, mood and stress easier to review.	Supports awareness and reflection; it does not diagnose or treat conditions.
SDG 10 · Reduced Inequalities	A readable interface and focus on free habits aim to make wellness information more accessible.	An inclusion goal, not proven impact; accessibility and user research still matter.

Presentation summary

Our strongest connection is SDG 3 because the app helps people reflect on everyday well-being habits. We also connect it to SDG 10 through the goal of making information understandable and accessible. Measuring social impact would require further research.

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09 Testing and what was verified

The team reports testing the project multiple times and creating a seven-day log to check the workflow, identify problems and fix them. This is a team-reported result and is not an independent test run for this document.

Suggested test matrix

Test	Expected behaviour
Valid record	Accepted and visible in records/dashboard.
Negative sleep or activity	Rejected with a clear validation message.
Water is zero or negative	Rejected.
Mood or stress outside its scale	Rejected when supplied.
Duplicate date	Rejected without replacing the earlier entry.
Too few days for a trend	Shows insufficient data instead of making a trend claim.
CSV and JSON export	Contains the expected fields and report values.
Normal user opens admin-only route	Denied or redirected according to route policy.
Empty account report	Shows a useful empty state without a server error.

10 Limitations and next steps

Current limitations

- Entries are manual and may be incomplete or inaccurate.
- Scoring uses fixed rules and is not clinically validated.
- The chatbot answers only supported questions.
- Local SQLite storage limits large-scale concurrent use.
- Accessibility, privacy and performance need more testing.

Future improvements

- Explore optional device syncing and more adaptive analysis.
- Improve the assistant and explain recommendations more clearly.
- Strengthen deployment, privacy controls and testing.
- Improve accessibility, language support and low-bandwidth use.

11 Team credits

Documentation: Swarit and Yunay

Coding and demonstration: Jivraj and Vaibhav

Presentation: Venisha and Pradishan

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12 Conclusion

HealthBridge brings everyday wellness habits into one accessible dashboard and turns daily entries into a simple score, trend summaries and practical suggestions. Its layered design, validation rules and OOP structure make the project useful for demonstrating web development and data processing. The score and recommendations are intended for awareness, not medical diagnosis. With stronger testing, accessibility and privacy safeguards, the project could become more reliable and inclusive.

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